Waste Management and Recycling System

Yıldız İşleyen

I. Introduction

In this study, a database application was developed that aims to provide an efficient process for waste management and recycling system.

The system was planned in accordance with the user types:

Citizens using the system can report information about the waste they left at the collection point through the system. Citizens can also view historical waste informations from their own profile. Another feature that the system provides for citizens is profile updating. Citizens can edit the password they created during registration from the profile section. Citizens can delete their account in the system if they wish.

Workers can see the information about the waste at the point they are assigned to and follow their task list.

Municipality officials can follow all the waste in their region through the system, while the citizen's personal information remains anonymous, and can see the workers assigned to their region. Officers can also remove workers via their own panel. In addition, they can also examine analyses such as daily or monthly total waste amount via their own panel.

II. Database Design and Development

A. Requirement analysis

The waste management and recycling system has been developed to ensure that all processes are monitored more conveniently and that the collected data is kept in a central system. The main functionalities are as described below.

- User registration
  Users must first register to log in to the system. They cannot register again with an existing e-mail address. During registration, the user enters their name, surname, e-mail, password and user type.
- Authorization according to different user types
  User can choose user type during registration. There are three different user types; citizen, worker, official. Each has its own panels. However, this feature may be changed in future developments as it may cause security vulnerabilities.
- Waste management
  Citizens can view past waste records from their own panel. They can make new waste notifications. When making a waste notification, weight and type of the waste, whether plastic, glass or organic, must be specified. The waste they report in the region they choose can be viewed by the worker assigned to that point. Workers do this from their own panel.
- Collection point management
  There are two different regions in the system. There are 2 collection points in these regions. This number can be increased in future developments. The system can be expanded. Each of these regions is assigned to a different officer. Each region in the system is assigned to only one officer. However, there can be more than one worker in a region. Officers can access information about their region from their own panel. They can see the workers and waste history in that region.
- User security
  Citizens can permanently delete their accounts and update their passwords at any time. While officers and workers view waste history, citizen information remains anonymous. Citizen personal information is not shared.

B. E/R Design

As seen in Figure 1, there are 6 different entities. These are User, Waste, CollectionPoint, RecyclingResult, WorkerAssignment and ManagerRegion. The relationships between them are established in accordance with the system. The relationship between User and Waste is 1 to many because a citizen can enter more than one waste record, and more than one waste can belong to a single user. The relationship between CollectionPoint and Waste is 1 to many because a collection point can contain more than one waste. The relationship between Waste and RecyclingResult (which keeps track of how much of the waste is recycled) is 1 to 1 because each waste record can only have one recycling result. The relationship between ManagerRegion and User is 1 to 1 because an officer is assigned to only one region.

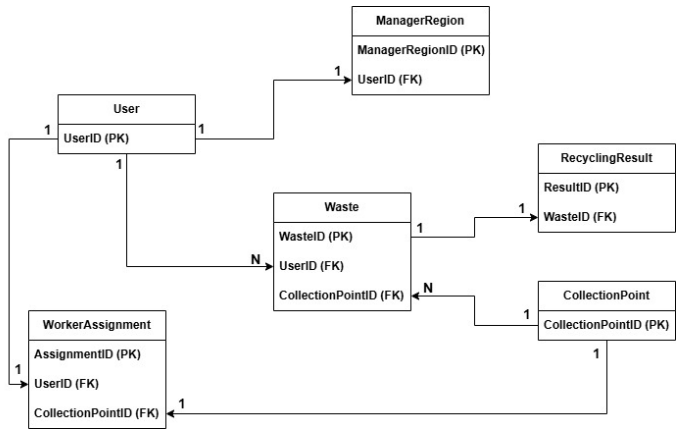


Fig. 1. E/R Design of Waste Management and Recycling System

C. Relational Database Schema

MySQL was used as the database in this project. There are 6 tables in the database. Each table has its own special attributes. Foreign keys are referenced in the tables established in pairs.

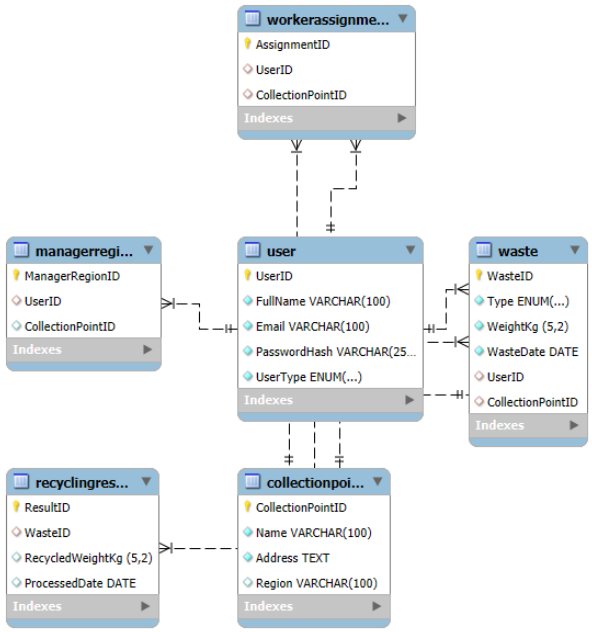


Fig. 2. Relational Database Schema

User: Keeps user information. (UserID, FullName, Email, PasswordHash, UserType)

Waste: Keeps waste information. (WasteID, Type, WeightKg, WasteDate, UserID, CollectionPointID)

CollectionPoint: Keeps waste collection point information. (CollectionPointID, Name, Address, Region)

Recycling Result: Keeps information on how much of the waste is recycled. (ResultID, RecycledWeightKg, ProcessedDate, WasteID)

WorkerAssignment: Keeps location information where workers are assigned. (AssignmentID, UserID, CollectionPointID)

ManagerRegion: Keeps region information where the officer is assigned. (ManagerRegionID, UserID)

III. Application Development

The technologies and tools used in this project were brought together to develop a web-based recycling tracking system. Python is used as the basic programming language of the application, while the Flask framework provides management of HTTP requests, routing operations and the establishment of the general web infrastructure. The necessary information for the system, such as user and waste, is securely stored with the MySQL database. The notification messages to the users are returned with Flash.



Fig. 3. Index Page

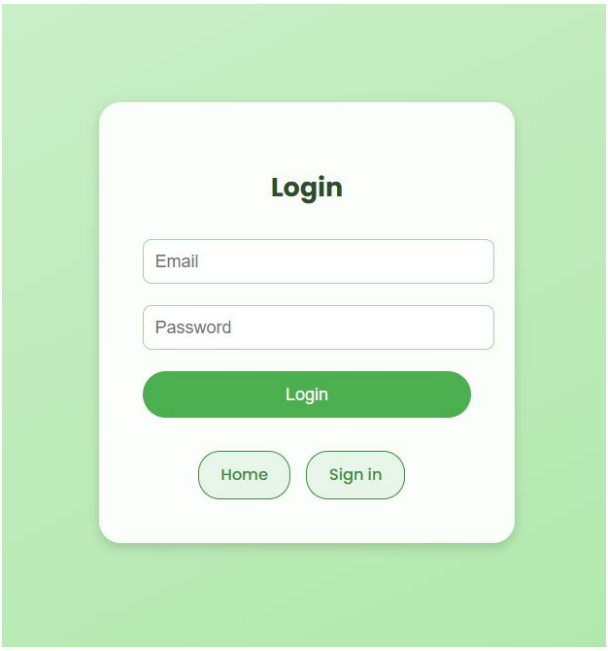


Fig. 4. Login Page

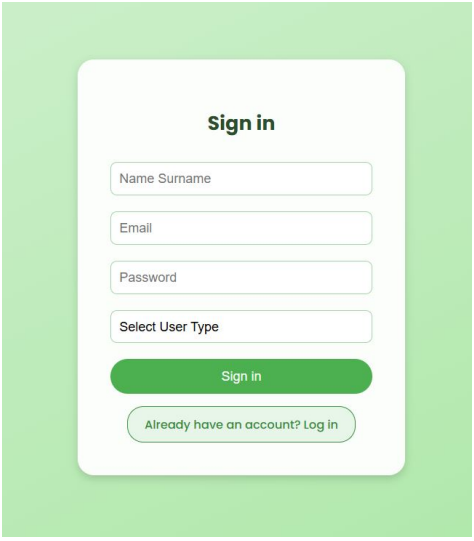


Fig. 5. Register Page

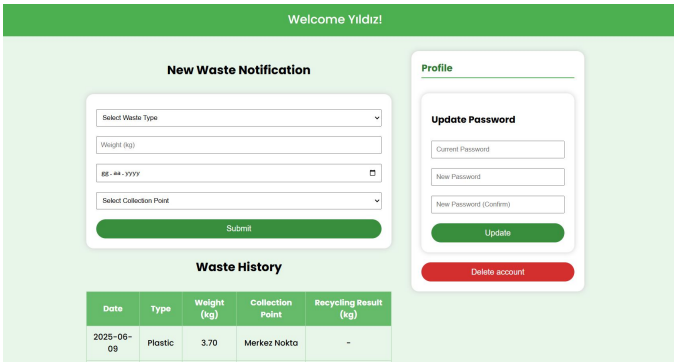


Fig. 6. Citizen Dashboard

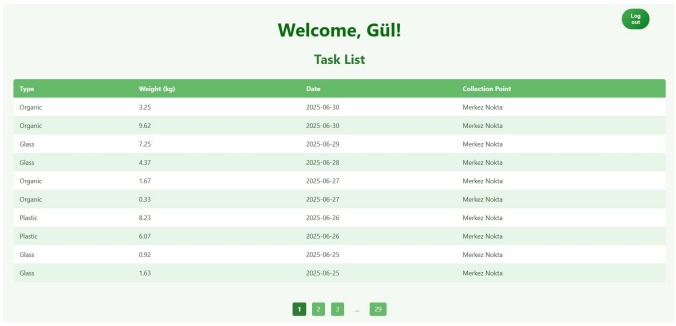


Fig. 7. Worker Dashboard

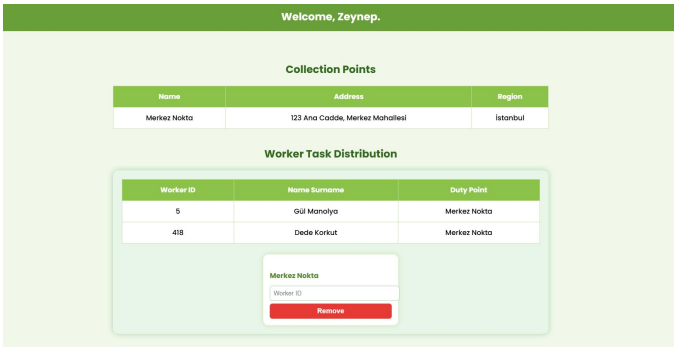


Fig. 8. Official Dashboard First Part



Fig. 9. Official Dashboard Second Part

IV. Challenges and Improvements

The main challenge was that each officer can only be assigned to one region. The authorized officer of the region should only be able to see the waste and workers in that region. In addition, the officer has the opportunity to remove the worker in that region. This problem was solved by creating a separate database table, ManagerRegion. The reason for this is to manage the regions more easily and to reduce information pollution.

The project analyzes the total daily/monthly waste amount. It is displayed on the officer page.

In the future, these analyzes can be differentiated, such as total weight according to waste type, users who recycle the most monthly. With this information, the municipality can make data-based decisions. Sustainability can be achieved by conducting comprehensive analyzes.

Another development could be to add supporting texts to the recycling citizen that will be visible on their profile.

V. Conclusion

In this project, waste management and recycling system was made by taking into account the types of users. With this study, each user using the system benefits from different aspects. Recycling has been facilitated and encouraged for the citizen. The interaction between the worker and the officer has been improved. Users have been provided ease of use with the panels provided. With this study, a system that values the environment has been developed.

References

[1] Flask. Session Documentation. https://flask.palletsprojects.com/en/stable/quickstart/#sessions

[2] MySQL. MySQL Documentation. https://dev.mysql.com/doc/